CHAPTER 1

Rong
Tang

Yun
Yang

2019–
2025

An-
nals
of
Sta-
tis-
tics
JRSS
B
AIS-
TATS
ICML

MCMC

1.

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$d \ll$
D

D

GAN
VAE

(1)

(2)

(3)

MCMC

Tang-
Yang

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2.

2.1. ?]]tang2022minimax

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Rong Tang and Yun Yang. “Min-i-max

Rate of Distribution Estimation on Unknown Submanifold under Adversarial Losses." *Annals of Statistics*, 2022.

2.1.1. *Summary.*

**ad-
ver-
sar-
ial
loss**

**min-
i-
max**

$$n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\gamma\beta}{d}},$$

d

α

β

γ

d

D

Fano

Le Cam

chaining

Caffarelli

2.1.2. *Overall Assessment.*

γ

1

d

2

**Lep-
ski**

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”

”

3

?]]tang2023lowdim
?]]tang2024diffusion
?]]tang2025regression

2.2. ?]]tang2021vae

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VAE

Excess Risk Bound.

Rong Tang and Yun Yang. "Excess Risk Bounds for Distribution Estimation with VAE." *COLT*, 2021.

VAE

encoder-decoder

VAE

ex-
cess
risk

VAE

1

2
VAE

pos-
te-
rior
col-
lapse

3

ex-
cess
risk
bound

2.3. ?]]tang2023robust

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Rong Tang, Anirban Bhattacharya, Debdeep Pati, and Yun Yang. "Robust Bayesian Inference on Riemannian Submanifold," *JRSS B*, 2023.

Bernstein-von Mises

(BvM)

**set-
ting**

**Me-
trop-
o-
lis
(RRWM)**

d

D

**con-
duc-
tance
pro-
file**

1

C^3

2
BvM

3
RRWM

pro-
posal

2.4. ?]]tang2024mala — Metropolis-Adjusted Langevin

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Rong Tang and Yun Yang. "Metropolis-Adjusted Langevin Algorithm: Optimal Dimension

Dependency and Beyond." *ICML*, 2024.

s-conductance profile

conductance

Chen et al. (2020)

conductance profile

warm-start

$\log M_0$

$\log \log M_0$

M_0

MALA

$O(d^{1/3})$

burn-in

Gibbs

1

$d \leq$
$cn^{\kappa_1}/\log n$

$d^{1/3}$

2
s-
warm
start

MCMC
”

”

3

burn-
in

?]]tang2023robust

?]]tang2023robust

BvM

MALA

2.5. ?]]tang2023twosample

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Rong Tang and Yun Yang. “Minimax Nonparametric Two-Sample Test

**un-
der
Ad-
ver-
sar-
ial
Losses,"
*AIS-
TATS*,
2023.**

Besov

1

$O(n \log n +$
$n^{2d/(4\alpha+d)})$

MMD

$O(n^2)$

1

2

J

α

α

3 Bootstrap

2.6. ?]]tang2024diffusion

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Rong Tang and Yun Yang. "Adaptivity of Diffusion

**Mod-
els
to
Man-
i-
fold
Struc-
tures," *AIS-
TATS*, 2024.**
Langevin

**Gauss-
ian**

$p_\sigma(x) = \int p_{\text{data}}(y)\phi_\sigma(x - y)\,\mathrm{d}y$

$t > 0$

Lebesgue

$\tilde{O}(\sigma)$

L_∞

-

1-
Wasserstein

$$\frac{1}{\sqrt{n}} \vee n^{-\frac{\alpha+1}{2\alpha+d}}.$$

1
Langevin

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2

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3

J

α

?]]tang2022minimax

?]]tang2022minimax

$\gamma =$
1

**Wasser-
stein**

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$\Leftrightarrow$

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2.7. ?]]tang2023lowdim

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Rong Tang and Yun Yang. “Estimating Distributions with Low-dimensional Structures

**Us-
ing
Mix-
tures
of
Gen-
er-
a-
tive
Mod-
els.”
ICML,
2023.**

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**MLD-
MAE
Mix-
ture
of**

**La-
tent
Dis-
tri-
bu-
tion
Matched
Auto-
Encoders**

K

-

ρ_k

**soft-
max**

**1-
Wasserstein**

1

K

$K \approx$
10

MNIST

2

$\alpha >$
$\beta -$
1

3

2.8. ?]]tang2025regression

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Rong Tang and Yun Yang.

“Minimax Optimal Rates for Regression on Manifolds and Distributions,” arXiv, 2025.

IPM

γ-
Hölder
IPM
d_γ

$\gamma =$
0

1-
Wasserstein

$\gamma =$
1

Regime

$\frac{\alpha_Y}{\alpha_X}$.
d_X

ad
hoc

$\gamma \neq$
$0, 1$

IPM

2.9. ?]]tang2025conditional

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Rong Tang and Yun Yang. “Conditional Diffusion Models are Minimax-Optimal and Manifold-Adaptive,” arXiv, 2025.

(d_X, d_Y)

(D_X, D_Y)

1

?]]tang2025regression

2

classifier-free guidance

3

3.

3.1.

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Tang-
Yang

(1)

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D

(2)

**Gir-
sanov**

s**-
conductance
pro-
file**

(3)

$\rightarrow$

VAE

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3.2.

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(1)

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d

(2)

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**de-
con-
vo-
lu-
tion**

(3)

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(4)

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(5)

β-

$\beta \geq$

1

4.

(a)

2021

—

?]]tang2021vae
VAE

(b)

2022

—

?]]tang2022minimax

(c)
2023a
—
?]]tang2023lowdim
MLD-
MAE
(d)
2023b
—
?]]tang2023robust

+
BvM
(e)
2023c
—
?]]tang2023twosample

(f)
2024a
—
?]]tang2024diffusion

(g)

2024b

—

?]]tang2024mala
MALA

(h)

2025a

—

?]]tang2025regression

(i)

2025b

—

?]]tang2025conditional

?]]**tang2021vae**
$\longrightarrow$
?]]$tang2024diffusion$ $\longrightarrow$
?]]$tang2025conditional$

?]]**tang2023robust**
$\xrightarrow{\textbf{BvM}}$
?]]$tang2024mala$

5.

(i)

Partition of Unity
?]]**tang2022minimax**
?]]**tang2023lowdim**

(ii)

γ-
Hölder
IPM
?]]tang2022minimax
?]]tang2023twosample
?]]tang2025regression

(iii)

Gaussian

$+$

?]]tang2024diffusion

Langevin

(iv)

?]]tang2024diffusion
?]]tang2025conditional

(v)
s-
conductance

**pro-
file
?]]tang2024mala**

**con-
duc-
tance**

**con-
duc-
tance
pro-
file**

(vi)

Bernstein-von Mises

?]]tang2023robust

BvM

setting

(vii) Girsanov

?]]tang2024diffusion
?]]tang2025conditional